

Arithmetic Logic Unit

The lab's goal is to construct an Arithmetic Logic Unit (ALU). An ALU is a combinational circuit that can perform mathematical operations and is a necessary building block for many digital devices.

Process

8 function operators are chosen to be defined in this module. Functions are:

1. Adder
2. Subtractor
3. AND
4. OR
5. XOR
6. NOT
7. Shift Left
8. Shift Right

A separate module is defined for each function. In the top module, all modules are called and ALU is made. Schematic and Basys images were taken in the laboratory. The code was rewritten because of a system error on the computer.

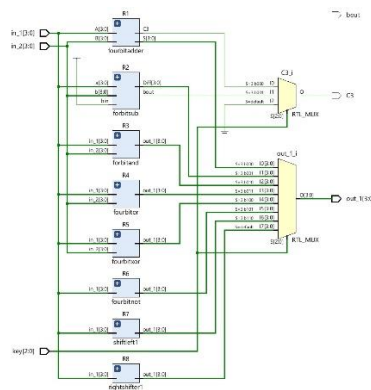


Figure 1: Schematic of the design

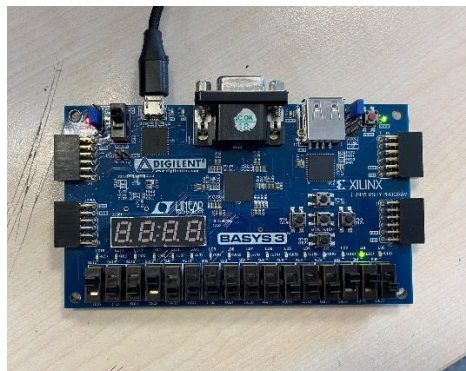


Figure 2: Shift_left operator

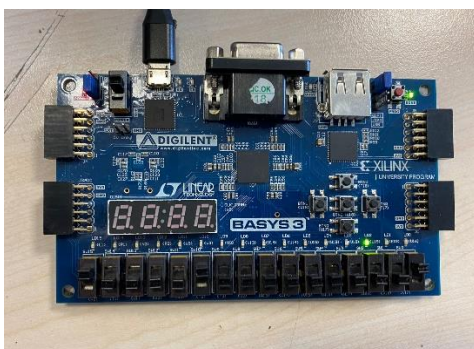


Figure 3: f_bitsubt operator

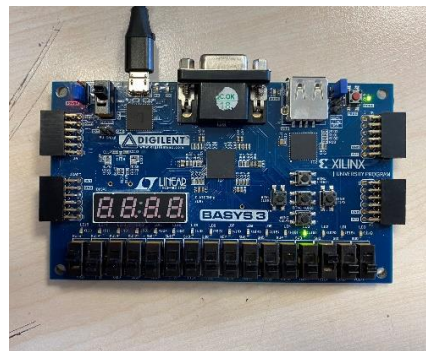


Figure 4: f_bitor operator

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Codes

```
library IEEE;

use IEEE.STD_LOGIC_1164.ALL;

entity f_bitadder is

    Port ( A : in STD_LOGIC_VECTOR (3 downto 0);
          B : in STD_LOGIC_VECTOR (3 downto 0);
          S : out STD_LOGIC_VECTOR (3 downto 0);
          C3 : out STD_LOGIC);

end f_bitadder;

architecture Behavioral of f_bitadder is
    COMPONENT fulladder is
        Port ( in_1 : in STD_LOGIC;
              in_2 : in STD_LOGIC;
              c_1 : in STD_LOGIC;
              sum_1 : out STD_LOGIC;
              c_o : out STD_LOGIC);
    end COMPONENT;
    signal C0,C1,C2: std_logic ;

begin

    FA1: fulladder port map (A(0),B(0),'0',S(0),C0) ;
    FA2: fulladder port map (A(1),B(1),C0,S(1),C1) ;
    FA3: fulladder port map (A(2),B(2),C1,S(2),C2) ;
    FA4: fulladder port map (A(3),B(3),C2,S(3),C3) ;
end Behavioral;

library IEEE;

use IEEE.STD_LOGIC_1164.ALL;

entity f_bitsubt is

    Port ( a : in STD_LOGIC_VECTOR (3
```

```
downto 0);

    b_in : in STD_LOGIC;

    Diff : out STD_LOGIC_VECTOR (3
downto 0);

    b_out : out STD_LOGIC);

end f_bitsubt;

architecture Behavioral of f_bitsubt is
    COMPONENT fulladder
    Port ( in_1 : in STD_LOGIC;
          in_2 : in STD_LOGIC;
          c_1 : in STD_LOGIC;
          sum_1 : out STD_LOGIC;
          c_o : out STD_LOGIC);
    end COMPONENT;
    signal sig: std_logic_vector(3 downto 0);
    signal s: std_logic_vector(3 downto 0);
    begin

        s <= not b;

        u1: fulladder port map(a(0), s(0), '1',
            Diff(0), sig(0));

        u2: fulladder port map(a(1), s(1), sig(0),
            Diff(1), sig(1));

        u3: fulladder port map(a(2), s(2), sig(1),
            Diff(2), sig(2));

        u4: fulladder port map(a(3), s(3), sig(2),
            Diff(3), sig(3));

    end Behavioral;
```

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```
library IEEE;  
use IEEE.STD_LOGIC_1164.ALL;
```

entity f_bitand is

```
    Port ( in_1 : in STD_LOGIC_VECTOR (3  
downto 0);  
          in_2 : in STD_LOGIC_VECTOR (3  
downto 0);  
          out_1 : out STD_LOGIC_VECTOR (3  
downto 0));  
end f_bitand;
```

architecture Behavioral of f_bitand is

COMPONENT and2

```
Port ( in_1 : in STD_LOGIC;  
      in_2 : in STD_LOGIC;  
      out_1 : out STD_LOGIC);  
end COMPONENT;
```

begin

```
A1: and2 port  
map(in_1(0),in_2(0),out_1(0));  
A2: and2 port  
map(in_1(1),in_2(1),out_1(1));  
A3: and2 port  
map(in_1(2),in_2(2),out_1(2));  
A4: and2 port  
map(in_1(3),in_2(3),out_1(3));
```

end Behavioral;

```
library IEEE;  
use IEEE.STD_LOGIC_1164.ALL;
```

entity f_bitor is

```
    Port ( in_1 : in STD_LOGIC_VECTOR (3  
downto 0);  
          in_2 : in STD_LOGIC_VECTOR (3  
downto 0);  
          out_1 : out STD_LOGIC_VECTOR (3  
downto 0));  
end f_bitor;
```

architecture Behavioral of f_bitor is

COMPONENT or2

```
Port ( in_1 : in STD_LOGIC;  
      in_2 : in STD_LOGIC;  
      out_1 : out STD_LOGIC);  
end COMPONENT;
```

begin

```
O1: or2 port  
map(in_1(0),in_2(0),out_1(0));  
O2: or2 port  
map(in_1(1),in_2(1),out_1(1));  
O3: or2 port  
map(in_1(2),in_2(2),out_1(2));  
O4: or2 port  
map(in_1(3),in_2(3),out_1(3));  
end Behavioral;
```

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```
library IEEE;  
use IEEE.STD_LOGIC_1164.ALL;
```

```
entity f_bitxor is  
    Port ( in_1 : in STD_LOGIC_VECTOR (3  
downto 0);  
          in_2 : in STD_LOGIC_VECTOR (3  
downto 0);  
          out_1 : out STD_LOGIC_VECTOR (3  
downto 0));  
end f_bitxor;
```

```
architecture Behavioral of f_bitxor is  
    COMPONENT xor2  
    Port ( in_1 : in STD_LOGIC;  
          in_2 : in STD_LOGIC;  
          out_1 : out STD_LOGIC);  
end COMPONENT;
```

```
begin  
X1: xor2 port  
map(in_1(0),in_2(0),out_1(0));  
X2: xor2 port  
map(in_1(1),in_2(1),out_1(1));  
X3: xor2 port  
map(in_1(2),in_2(2),out_1(2));  
X4: xor2 port  
map(in_1(3),in_2(3),out_1(3));
```

```
end Behavioral;
```

```
library IEEE;  
use IEEE.STD_LOGIC_1164.ALL;  
  
entity f_bitnot is  
    Port ( in_1 : in STD_LOGIC_VECTOR (3  
downto 0);  
          out_1 : out STD_LOGIC_VECTOR (3  
downto 0));  
end f_bitnot;
```

```
architecture Behavioral of f_bitnot is  
    COMPONENT not2  
    Port ( in_1 : in STD_LOGIC;  
          out_1 : out STD_LOGIC);  
end COMPONENT;
```

```
begin  
N1: not2 port map(in_1(0),out_1(0));  
N2: not2 port map(in_1(1),out_1(1));  
N3: not2 port map(in_1(2),out_1(2));  
N4: not2 port map(in_1(3),out_1(3));
```

```
end Behavioral;
```

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```
library IEEE;

use IEEE.STD_LOGIC_1164.ALL;

entity shift_left is

    Port ( in_1 : in STD_LOGIC_VECTOR (3
downto 0);

        out_1 : out STD_LOGIC_VECTOR (3
downto 0));

end shift_left;

architecture Behavioral of shift_left is

begin

out_1(3 downto 0) <= in_1(2 downto 0) &
'0';

end Behavioral;

library IEEE;

use IEEE.STD_LOGIC_1164.ALL;

entity shift_right is

    Port ( in_1 : in STD_LOGIC_VECTOR (3
downto 0);

        out_1 : out STD_LOGIC_VECTOR (3
downto 0));

end shift_right;

architecture Behavioral of shift_right is

begin

out_1(3 downto 0) <= '0' & in_1(3 downto
1);

end Behavioral;
```

```
library IEEE;

use IEEE.STD_LOGIC_1164.ALL;

entity fullsub is

    Port ( a : in STD_LOGIC;

        b : in STD_LOGIC;

        b_in : in STD_LOGIC;

        Diff : out STD_LOGIC;

        b_out : out STD_LOGIC);

end fullsub;

architecture Behavioral of fullsub is

begin

Diff <= a xor b xor b_in;

b_out <= (not a and b_in) or (not a and b)
or ( b and b_in);

end Behavioral;

library IEEE;

use IEEE.STD_LOGIC_1164.ALL;

entity ALU is

    Port ( in_1 : in STD_LOGIC_VECTOR (3
downto 0);

        in_2 : in STD_LOGIC_VECTOR (3
downto 0);

        key : in STD_LOGIC_VECTOR (2
downto 0);

        out_1 : out STD_LOGIC_VECTOR (3
downto 0);
```

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```
        C3,b_out : out std_logic);  
end ALU;
```

architecture Behavioral of ALU is

```
COMPONENT f_bitadder
```

```
    Port ( A : in STD_LOGIC_VECTOR (3  
downnto 0);
```

```
        B : in STD_LOGIC_VECTOR (3  
downnto 0);
```

```
        S : out STD_LOGIC_VECTOR (3  
downnto 0);
```

```
        C3 : out STD_LOGIC);  
end COMPONENT;
```

```
COMPONENT f_bitsubt
```

```
    Port ( a : in STD_LOGIC_VECTOR (3  
downnto 0);
```

```
        b : in STD_LOGIC_VECTOR (3  
downnto 0);
```

```
        b_in : in STD_LOGIC;  
        Diff : out STD_LOGIC_VECTOR (3  
downnto 0);
```

```
        b_out : out STD_LOGIC);  
end COMPONENT;
```

```
COMPONENT f_bitand
```

```
    Port ( in_1 : in STD_LOGIC_VECTOR (3  
downnto 0);
```

```
        in_2 : in STD_LOGIC_VECTOR (3  
downnto 0);
```

```
        out_1 : out STD_LOGIC_VECTOR (3  
downnto 0));
```

```
end COMPONENT;
```

```
COMPONENT f_bitor
```

```
    Port ( in_1 : in STD_LOGIC_VECTOR (3  
downnto 0);
```

```
        in_2 : in STD_LOGIC_VECTOR (3  
downnto 0);
```

```
        out_1 : out STD_LOGIC_VECTOR (3  
downnto 0));  
end COMPONENT;
```

```
COMPONENT f_bitxor
```

```
    Port ( in_1 : in STD_LOGIC_VECTOR (3  
downnto 0);
```

```
        in_2 : in STD_LOGIC_VECTOR (3  
downnto 0);
```

```
        out_1 : out STD_LOGIC_VECTOR (3  
downnto 0));  
end COMPONENT;
```

```
COMPONENT f_bitnot
```

```
    Port ( in_1 : in STD_LOGIC_VECTOR (3  
downnto 0);
```

```
        out_1 : out STD_LOGIC_VECTOR (3  
downnto 0));  
end COMPONENT;
```

```
COMPONENT shift_left
```

```
    Port ( in_1 : in STD_LOGIC_VECTOR (3  
downnto 0);
```

```
        out_1 : out STD_LOGIC_VECTOR (3  
downnto 0));  
end COMPONENT;
```

COMPONENT shift_right

Port (in_1 : in STD_LOGIC_VECTOR (3
downto 0);

out_1 : out STD_LOGIC_VECTOR (3
downto 0));

end COMPONENT;

signal sig0,sig1,sig2,sig3,sig4,sig5,sig6,sig7:
std_logic_vector(3 downto 0);

signal sig8,sig9,sig10,sig11: std_logic;

begin

R1: f_bitadder port library IEEE;

use IEEE.STD_LOGIC_1164.ALL;

entity tb is

Port ();

end tb;

architecture Behavioral of tb is

COMPONENT ALU

Port (in_1 : in STD_LOGIC_VECTOR (3
downto 0);

in_2 : in STD_LOGIC_VECTOR (3
downto 0);

key : in STD_LOGIC_VECTOR (2
downto 0);

out_1 : out STD_LOGIC_VECTOR (3
downto 0);

C3, b_out: out std_logic);

end COMPONENT;

signal in_1: std_logic_vector(3 downto 0);

signal in_2: std_logic_vector(3 downto 0);

signal key: std_logic_vector(2 downto 0);

signal out_1: std_logic_vector(3 downto
0);

signal C3: std_logic;

signal b_out: std_logic;

begin

uut: ALU port map(

in_1=>in_1,

in_2=>in_2,

key=>key,

out_1=>out_1,

C3=>C3,

b_out=>b_out);

tb:process

begin

wait for 50 ns;

key<= "000";

in_1<="1101";

in_2<="1000";

wait for 50 ns;

key<= "001";

in_1<="1101";

in_2<="1000";

wait for 50 ns;

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```
key<= "010";
in_1<="1101";
in_2<="1000";
wait for 50 ns;
key<= "011";
in_1<="1101";
in_2<="1000";
wait for 50 ns;
key<= "100";
in_1<="1101";
in_2<="1000";
wait for 50 ns;
key<= "101";
in_1<="1101";
in_2<="1000";
wait for 50 ns;
key<= "110";
in_1<="1101";
in_2<="1000";
wait for 50 ns;
key<= "111";
in_1<="1101";
in_2<="1000";
wait for 50 ns;
key<= "000";
in_1<="0101";
in_2<="0011";
wait for 50 ns;
key<= "001";

in_1<="0101";
in_2<="0011";
wait for 50 ns;
key<= "010";
in_1<="0101";
in_2<="0011";
wait for 50 ns;
key<= "011";
in_1<="0101";
in_2<="0011";
wait for 50 ns;
key<= "100";
in_1<="0101";
in_2<="0011";
wait for 50 ns;
key<= "101";
in_1<="0101";
in_2<="0011";
wait for 50 ns;
key<= "110";
in_1<="0101";
in_2<="0011";
wait for 50 ns;
key<= "111";
in_1<="0101";

end process;

end Behavioral;map(in_1,in_2,sig0,sig8);

R2: f_bitsubt port
map(in_1,in_2,'1',sig1,sig9);
```


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R3: f_bitand port map(in_1,in_2,sig2);

R4: f_bitor port map(in_1,in_2,sig3);

R5: f_bitxor port map(in_1,in_2,sig4);

R6: f_bitnot port map(in_1,sig5);

R7: shift_left port map(in_1,sig6);

R8: shift_right port map(in_1,sig7);

with key select

out_1 <= sig0 when "000",

sig1 when "001",

sig2 when "010",

sig3 when "011",

sig4 when "100",

sig5 when "101",

sig6 when "110",

sig7 when others;

with key select

C3 <= sig8 when "000",

sig9 when "001",

'0' when others;

end Behavioral;

set_property PACKAGE_PIN V17 [get_ports {key[0]]

set_property IOSTANDARD LVCMOS33 [get_ports {key[0]]

set_property PACKAGE_PIN V16 [get_ports {key[1]]

set_property IOSTANDARD LVCMOS33 [get_ports {key[1]]

set_property PACKAGE_PIN W16 [get_ports {key[2]]

set_property IOSTANDARD LVCMOS33 [get_ports {key[2]]

set_property PACKAGE_PIN V2 [get_ports {in_2[0]]

set_property IOSTANDARD LVCMOS33 [get_ports {in_2[0]]

set_property PACKAGE_PIN T3 [get_ports {in_2[1]]

set_property IOSTANDARD LVCMOS33 [get_ports {in_2[1]]

set_property PACKAGE_PIN T2 [get_ports {in_2[2]]

set_property IOSTANDARD LVCMOS33 [get_ports {in_2[2]]

set_property PACKAGE_PIN R3 [get_ports {in_2[3]]

set_property IOSTANDARD LVCMOS33 [get_ports {in_2[3]]

set_property PACKAGE_PIN W2 [get_ports {in_1[0]]

set_property IOSTANDARD LVCMOS33 [get_ports {in_1[0]]

set_property PACKAGE_PIN U1 [get_ports {in_1[1]]

set_property IOSTANDARD LVCMOS33 [get_ports {in_1[1]]

set_property PACKAGE_PIN T1 [get_ports {in_1[2]]

set_property IOSTANDARD LVCMOS33 [get_ports {in_1[2]]

set_property PACKAGE_PIN R2 [get_ports {in_1[3]]

set_property IOSTANDARD LVCMOS33 [get_ports {in_1[3]]

##leds

set_property PACKAGE_PIN U16 [get_ports {out_1[0]]

set_property IOSTANDARD LVCMOS33 [get_ports {out_1[0]]

set_property PACKAGE_PIN E19 [get_ports {out_1[1]]

set_property IOSTANDARD LVCMOS33 [get_ports {out_1[1]]

set_property PACKAGE_PIN U19 [get_ports {out_1[2]]

set_property IOSTANDARD LVCMOS33 [get_ports {out_1[2]]

set_property PACKAGE_PIN V19 [get_ports {out_1[3]]

set_property IOSTANDARD LVCMOS33 [get_ports {out_1[3]]

set_property PACKAGE_PIN U15 [get_ports {C3}]

set_property IOSTANDARD LVCMOS33 [get_ports {C3}]

set_property PACKAGE_PIN U14 [get_ports {b_out}]

set_property IOSTANDARD LVCMOS33 [get_ports {b_out}]